

Seminar 10 and 11 - 2026

Theoretical aspects

A sequence $(X_n)_{n \in \mathbb{N}}$ of random variables **converges in probability** to a random variable X , denoted by $X_n \xrightarrow{\mathbb{P}} X$, if

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \leq \varepsilon) = 1 \quad \text{for every } \varepsilon > 0.$$

A sequence $(X_n)_{n \geq 1}$ of random variables converges **in mean square** to a random variable X if

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - X|^2] = 0.$$

This convergence is denoted by $X_n \xrightarrow{L^2} X$.

A sequence $(X_n)_{n \geq 1}$ of random variables converges **in distribution** to a random variable X if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

in each continuity point x of F_X . This convergence is denoted by $X_n \xrightarrow{d} X$.

1. A program returns a value according to a random variable X with $E(X) = m \in \mathbb{R}$ and $V(X) = \sigma^2$, $\sigma > 0$. Prove that X takes values in the interval $(m - 3\sigma, m + 3\sigma)$ with more than 88% probability.
2. The number of items produced in a factory during a day is a random variable with mean 50. If we consider a day, which event is more likely: E_1 : "the production is more than 100 items in this day" or E_2 : "the production is at most 100 items in this day"?

3. Let $(X_n)_{n \in \mathbb{N}^*}$ be a sequence of independent random variables with $Unif[a, b]$ distribution, where $a < b$. Define for each $n \in \mathbb{N}^*$

$$Y_n = \max\{X_1, \dots, X_n\} \quad \text{and} \quad Z_n = \min\{X_1, \dots, X_n\}.$$

Prove that $Y_n \xrightarrow{P} b$ and $Z_n \xrightarrow{P} a$.

4. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of Bernoulli random variables. Prove that $X_n \xrightarrow{P} 0$ if and only if $X_n \xrightarrow{L^2} 0$.
5. Let $\lambda > 0$. A calling center has the following property, for every $n \in \mathbb{N}$, $n \geq 100$, during an hour interval $(0, 1]$: the calls arrive independently with at most one call in each time subinterval $(\frac{i}{n}, \frac{i+1}{n}]$, one call has probability $\frac{\lambda}{n}$ to occur, $i = \overline{0, n-1}$. Let's denote by X_n the corresponding total number of calls. Prove that $X_n \xrightarrow{d} X$, where $X \sim Poiss(\lambda)$.

6. Let $(X_n)_{n \in \mathbb{N}^*}$ be a sequence of independent random variables with $Unif[0, 1]$ distribution. Define for each $n \in \mathbb{N}^*$

$$Y_n = \max\{X_1, \dots, X_n\} \quad \text{and} \quad Z_n = \min\{X_1, \dots, X_n\}.$$

Prove that $Y_n \xrightarrow{L^2} 1$ and $Z_n \xrightarrow{L^2} 0$.

7. Consider a sequence of distinct coins such that the probability of getting a head with the n th coin is $\frac{1}{n}$, $n \in \mathbb{N}^*$. Let X_n be 1, if the toss of the n th coin shows a head, and 0, otherwise. Do we have $X_n \xrightarrow{a.s.} 0$?